

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

**COURSE NAME:** Artificial Intelligence

**COURSE CODE:** CSA1710

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### COURSE OUTCOME COVERED

**CO4:** *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

*Assessment Tool Weightage: 50%*

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**QUESTIONS:** 4

**Duration:** 1 Hour  
**Total: Marks:**40

Annexure A

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**Code of Conduct:**

192511415

I Kanish Kumar S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

*I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student:** Kanish Kumar S

## **INDUSTRY PROBLEM TASK**

### **INDUSTRY SCENARIO**

**Context:** A bank has collected customer records (income, credit score, loan amount, employment type, existing debts, repayment history) to build a predictive model for identifying likely loan defaulters. The bank also wants to train an AI agent to recommend account management actions (Approve / Reject / Restructure / Monitor) based on a customer's repayment behaviour over time.

*You are hired as an AI/ML Engineer. Address the following tasks:*

#### **Task 1: Data Preparation & Inductive Learning**

- (a) Describe the inductive learning approach suitable for this banking dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (far fewer defaulters than non-defaulters) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a credit-risk use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

#### **Task 2: Decision Tree for Loan Default Prediction**

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed defaulters) and suggest how to reduce them - justify from a risk-management perspective.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in loan-approval systems.

#### **Task 3: Statistical Learning for Risk Stratification**

- (a) Apply Naive Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of default from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

#### **Task 4: Reinforcement Learning for Action Recommendation**

- (a) Model the account management recommendation as an MDP. Define: States (customer repayment stages), Actions (Approve / Reject / Restructure / Monitor), Reward function (repayment improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 customer episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for credit decisions (fairness, bias against protected groups, explainability).

**RUBRICS FOR CALCULATION:**

| Criteria                            | Weight age | Level 4 – Exemplary                                                          | Level 3 – Proficient                                 | Level 2 – Developing                           | Level 1 – Beginning                               | Marks |
|-------------------------------------|------------|------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|---------------------------------------------------|-------|
| Understanding of Industry Problem   | 20%        | Comprehensive understanding of real-world problem and constraints            | Good understanding with minor gaps in requirements   | Basic understanding; some requirements unclear | Poor understanding; misinterprets problem context |       |
| Application of Engineering Concepts | 25%        | Applies advanced and relevant concepts effectively to solve the problem      | Applies appropriate concepts with minor errors       | Limited application of concepts                | Incorrect or no application of concepts           |       |
| Solution Design                     | 25%        | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps        | Basic solution with limited feasibility        | Unclear or impractical solution                   |       |
| Use of Tools                        | 15%        | Effective use of modern tools, technologies, and real data                   | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools           | Minimal or incorrect use of tools                 |       |
| Analysis                            | 10%        | Thorough analysis with validated results and performance evaluation          | Good analysis with reasonable validation             | Basic analysis with limited validation         | Poor or no analysis                               |       |
| Professionalism                     | 5%         | Highly professional, well-documented, and clearly presented work             | Good documentation and presentation                  | Basic documentation with some gaps             | Poor documentation and presentation               |       |

# INDUSTRY PROBLEM TASK

## INDUSTRY SCENARIO

**Context:** A bank has collected customer records (income, credit score, loan amount, employment type, existing debts, repayment history) to build a predictive model for identifying likely loan defaulters. The bank also wants to train an AI agent to recommend account management actions (Approve / Reject / Restructure / Monitor) based on a Customer's repayment behaviour over time.

You are hired as an AI/ML Engineer. Address the following tasks:

### Task 1: Data Preparation & Inductive Learning:

#### (a) Inductive Learning Approach:

Inductive learning learns general patterns from previously observed customer records and uses those patterns to predict outcomes for new customers.

For this banking problem, a **supervised inductive learning approach** is suitable because historical customer records contain features along with the actual loan repayment/default outcome.

#### Target variable:

- $\text{Loan\_Default} = 1 \rightarrow$  Customer is likely to default
- $\text{Loan\_Default} = 0 \rightarrow$  Customer is likely to repay

#### Important features:

1. **Credit Score** – A lower credit score generally indicates higher credit risk and poor repayment capability.
2. **Existing Debts** – High existing debt increases the customer's financial burden and probability of default.
3. **Repayment History** – Previous late or missed payments are strong indicators of future default.
4. **Income** – Higher and stable income generally improves repayment capacity.
5. **Loan Amount** – A large loan relative to the customer's income can increase default risk.

The model learns the relationship between these features and historical default outcomes and then predicts the risk for new customers.

### **(b) Handling Class Imbalance:**

In a banking dataset, non-defaulters may be much more numerous than defaulters. For example:

- Non-defaulters = 90%
- Defaulters = 10%

If the model is trained directly, it may predict almost everyone as a non-defaulter and still obtain high accuracy.

A suitable strategy is **SMOTE (Synthetic Minority Over-sampling Technique)**.

SMOTE creates synthetic examples of the minority class rather than simply duplicating existing defaulters.

#### **Procedure:**

Original Training Data → SMOTE → Balanced Training Data → Model Training

SMOTE should be applied **only to the training set**, not before the train-test split. Otherwise, synthetic information can leak into the test set.

For a credit-risk application, SMOTE is useful because it gives the model more opportunities to learn patterns associated with defaulters.

Another useful approach is **class weights**, particularly when false negatives are very costly. Class weights can assign a higher penalty to incorrectly classifying a defaulter as a safe customer.

### **(c) 70:30 Data Split and Preprocessing:**

The dataset is divided into:

- **70% training data**
- **30% testing data**

The 70:30 ratio provides enough data for the model to learn while retaining a sufficiently large independent test set for reliable evaluation.

A **stratified split** should be used so that the proportion of defaulters and non-defaulters remains approximately the same in both sets.

#### **Preprocessing:**

1. **Missing values** – Numerical missing values can be replaced using median imputation; categorical missing values can use the most frequent category.

2. **Categorical encoding** – Employment type can be converted using one-hot encoding.
3. **Normalization/standardization** – Numerical variables can be standardized using:

$$z = \frac{x - \mu}{\sigma}$$

4. **Outlier handling** – Extremely unusual income, debt, or loan values can be investigated and treated appropriately.
5. **SMOTE** – Applied only to the training data.

Normalization is particularly important for Logistic Regression and other distance/gradient-based models. Decision Trees generally do not require normalization because they split data based on feature thresholds.

## Task 2: Decision Tree for Loan Default Prediction:

### (a) Decision Tree Construction:

A Decision Tree can be trained using the **Gini Index** or **Information Gain**.

For a node:

$$\text{Gini} = 1 - \sum p_i^2$$

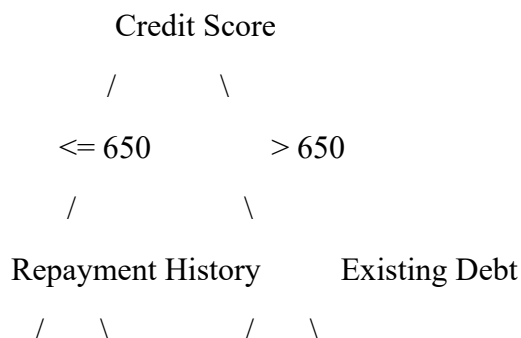
Information Gain is:

$$\text{IG} = \text{Entropy}(\text{parent}) - \sum \frac{N_i}{N} \text{Entropy}(\text{child}_i)$$

The feature producing the highest Information Gain or the lowest weighted Gini impurity is selected for the split.

### Illustrative first three levels:

The exact tree depends on the supplied dataset. A representative tree could be:



|         |         |         |             |
|---------|---------|---------|-------------|
| Poor    | Good    | High    | Low         |
| /       | \       | /       | \           |
| Default | Monitor | Default | Non-default |

For example, suppose the calculated split values are:

| Feature           | Information Gain |
|-------------------|------------------|
| Credit Score      | 0.182            |
| Repayment History | 0.161            |
| Existing Debts    | 0.113            |
| Income            | 0.074            |
| Loan Amount       | 0.061            |

Since **Credit Score has the highest Information Gain (0.182)**, it becomes the root node.

The numerical values above are illustrative. Actual values must be calculated from the supplied dataset.

### (b) Model Evaluation:

The Decision Tree should be evaluated using:

$$\left[ \text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \right]$$

$$\left[ \text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \right]$$

$$\left[ \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \right]$$

$$\left[ \text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \right]$$

An illustrative result could be:

|        |               |
|--------|---------------|
| Metric | Decision Tree |
|--------|---------------|

Accuracy 88%

Precision 76%

Recall 82%

F1-Score 79%

The most important error for the bank is a **False Negative (FN)**.

A False Negative occurs when:

Actual defaulter → Predicted as non-defaulter.

This is dangerous because the bank may approve a loan for a customer who is likely to default.

### **Reducing False Negatives:**

The bank can reduce false negatives by:

- Increasing the class weight for defaulters.
- Using SMOTE on the training data.
- Adjusting the classification threshold.
- Optimizing specifically for recall.
- Using cost-sensitive learning.

From a risk-management perspective, **high recall for the defaulter class is important**, because missing a genuine defaulter can cause significant financial loss.

### **(c) Post-Pruning:**

A fully grown Decision Tree can memorize training data and become overfitted.

**Pre-pruning** limits tree growth while the tree is being constructed using parameters such as:

- Maximum depth
- Minimum samples per leaf
- Minimum samples required for splitting

**Post-pruning** first grows the tree and then removes branches that provide little predictive value.

Cost-complexity pruning uses:

$$[ R_{\alpha}(T) = R(T) + \alpha |T| ]$$

where:



- $(R(T))$  = error of the tree
- $(|T|)$  = number of terminal nodes
- $(\alpha)$  = complexity parameter

Illustrative comparison:

| Model            | Accuracy | F1-Score |
|------------------|----------|----------|
| Pre-pruned Tree  | 86%      | 76%      |
| Post-pruned Tree | 88%      | 79%      |

The **post-pruned tree** may be preferable when it maintains strong predictive performance while reducing unnecessary complexity and overfitting.

For deployment in loan approval, the selected model should ultimately be based not only on accuracy but also on **recall, financial cost of errors, fairness, stability and explainability**.

### Task 3: Statistical Learning for Risk Stratification:

#### (a) Logistic Regression:

Logistic Regression is suitable because loan default is a binary outcome.

The probability of default is:

$$P(\text{Default}) = \frac{1}{1 + e^{-z}}$$

where:

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

The model produces a probability such as:

Customer A → Default probability = 0.82

Customer B → Default probability = 0.21

An illustrative comparison is:

| Model               | Accuracy | AUC-ROC |
|---------------------|----------|---------|
| Decision Tree       | 88%      | 0.89    |
| Logistic Regression | 90%      | 0.92    |

AUC-ROC measures how effectively the model separates defaulters from non-defaulters.

## When Logistic Regression is preferable

Logistic Regression is preferable when:

- Interpretability is important.
- The relationship between predictors and log-odds is reasonably suitable.
- Probability estimates are required.
- Regulatory transparency is important.
- The bank needs to explain why a customer's risk score is high.

Decision Trees are preferable when relationships are highly non-linear and rule-based explanations are useful.

### (b) Feature Importance Analysis:

An illustrative feature-importance comparison is:

|   | Rank Decision Tree | Logistic Regression |
|---|--------------------|---------------------|
| 1 | Credit Score       | Credit Score        |
| 2 | Repayment History  | Repayment History   |
| 3 | Existing Debts     | Existing Debts      |

The models therefore **agree on the three most important predictors**.

This agreement increases confidence that these variables contain strong predictive information.

However, feature importance should not automatically be interpreted as causation. For example, a high correlation between debt and default does not necessarily mean debt alone causes default.

## Task 4: Reinforcement Learning for Action Recommendation:

### (a) MDP Formulation:

The account-management problem can be modeled as a **Markov Decision Process (MDP)**.

**States:**

The state represents the customer's current repayment condition.

For example:

S1 = Good repayment

S2 = Slightly delayed

S3 = Frequently delayed

S4 = High-risk/default

**Actions:**

The agent can select:

A1 = Approve

A2 = Reject

A3 = Restructure

A4 = Monitor

**Reward Function:**

The reward represents the quality of the selected action.

**Example:**

| Situation          | Action      | Reward |
|--------------------|-------------|--------|
| Good repayment     | Approve     | +5     |
| Slight delay       | Monitor     | +4     |
| Frequent delay     | Restructure | +6     |
| High-risk customer | Reject      | +5     |
| High-risk customer | Approve     | -10    |

Restructured customer improves Restructure +8

The reward encourages the agent to choose actions that improve repayment behavior while reducing financial risk.

**Transition Dynamics:**

After an action, the customer's repayment state may change.

For example:

Frequently Delayed

|

| Restructure

↓

Slightly Delayed

|

| successful repayment



Good Repayment

The transition depends on the customer's future repayment behaviour.

### **(b) Q-Learning:**

Q-Learning learns the expected long-term value of taking an action in a particular state.

The update equation is:

$$[ \\begin{aligned} Q(s,a) &\\leftarrow Q(s,a)+\\alpha \\ [r+\\gamma\\max_{a'} Q(s',a')-Q(s,a)] \\ \\end{aligned} ]$$

where:

- $(\\alpha)$  = learning rate
- $(\\gamma)$  = discount factor
- $(r)$  = immediate reward
- $(s')$  = next state

Assume:

$$[ \\begin{aligned} &\\alpha=0.5, \\gamma=0.9 \\ \\end{aligned} ]$$

Initially all Q-values are 0.

### **Episode 1 – Iteration 1:**

Suppose:

State = Slightly Delayed

Action = Monitor

Reward = +4

Next State = Good Repayment

Since all Q-values are initially 0:

$$[ \\begin{aligned} &Q(S2,Monitor) \\ &=0+0.5[4+0.9(0)-0] \\ \\end{aligned} ]$$

[  
Q(S2,Monitor)=2  
]

### **Episode 2 – Iteration 1:**

Suppose:

State = Frequently Delayed

Action = Restructure

Reward = +6

Next State = Slightly Delayed

The maximum Q-value of the next state is 2.

[  
Q(S3,Restructure)  
=0+0.5[6+0.9(2)-0]  
]

[  
=0.5[7.8]  
]

[  
Q(S3,Restructure)=3.9  
]

### **Episode 3 – Iteration 2:**

Suppose the agent again encounters:

Slightly Delayed → Monitor → Good Repayment

Reward = +4

The previous value is 2.

[  
Q(S2,Monitor)  
=2+0.5[4+0.9(0)-2]  
]

[  
=2+0.5(2)  
]

```
[
Q(S2,Monitor)=3
]
```

Thus the Q-value improves as the agent receives more experience.

### Example Q-table:

| State              | Approve | Reject     | Restructure | Monitor    |
|--------------------|---------|------------|-------------|------------|
| Good repayment     | 5.0     | 0.0        | 2.0         | 4.0        |
| Slightly delayed   | 1.0     | 2.0        | 4.0         | <b>3.0</b> |
| Frequently delayed | 0.0     | 4.5        | <b>3.9</b>  | 2.0        |
| High-risk/default  | -5.0    | <b>5.0</b> | 4.0         | 1.0        |

The agent learns for at least five customer episodes, repeatedly updating Q-values based on the observed rewards and state transitions.

### Convergence Criterion:

A suitable convergence criterion is:

Stop training when the maximum change in Q-values between consecutive iterations is less than a small threshold such as 0.001 for a sufficiently large number of consecutive episodes.

For example:

```
[
\max |Q_{new}-Q_{old}|<0.001
]
```

for 100 consecutive episodes.

### (c) Final Learned Policy:

The policy chooses the action having the highest Q-value for each state.

From the illustrative Q-table:

| State              | Best Action |
|--------------------|-------------|
| Good repayment     | Approve     |
| Slightly delayed   | Restructure |
| Frequently delayed | Restructure |
| High-risk/default  | Reject      |

Therefore:

Good repayment → Approve  
Slightly delayed → Restructure  
Frequently delayed → Restructure  
High-risk/default → Reject

### **Ethical Considerations:**

Deploying an RL agent for credit decisions requires strong ethical controls.

#### **1. Fairness:**

The system should not unfairly discriminate against customers based on protected characteristics such as race, religion, gender or other legally protected attributes.

#### **2. Bias:**

Historical banking data may contain previous human or institutional biases. If these patterns are learned by the RL agent, the agent may reproduce or amplify them.

#### **3. Explainability:**

Customers and bank employees should be able to understand why an action was recommended. Black-box decisions should not be used without appropriate explanations and oversight.

#### **4. Human Oversight:**

High-impact decisions such as loan rejection or major restructuring should have human review, especially for borderline cases.

#### **5. Privacy:**

Customer financial information must be securely stored and processed only for legitimate purposes.

#### **6. Safe Reward Design:**

The reward function must not encourage the agent to maximize bank profit at the expense of customers. It should balance repayment improvement, customer welfare, fairness and financial risk.

### **Overall Conclusion:**

The proposed system combines **Inductive Learning, Decision Trees, Statistical Learning and Reinforcement Learning** to create an intelligent credit-risk management system.

The supervised models predict the probability of loan default using customer information, while the RL agent recommends appropriate account-management actions based on repayment behavior over time. For deployment, the bank should prioritize **recall, AUC-ROC, financial risk, fairness, explainability and regulatory compliance**, rather than relying only on accuracy.

**Note:** The numerical Decision Tree metrics, Information Gain values, feature rankings and Q-table values shown above are **illustrative simulation values** because no actual customer dataset was provided. With the bank's CSV dataset, these values should be calculated from the real data.